Code :

```
#include <iostream>
using namespace std;
char board[3][3] = {
   {'1','2','3'},
   {'4','5','6'},
   {'7','8','9'}
};
void displayBoard() {
   cout << "\n";
   for(int i = 0; i < 3; i++) {
      for(int j = 0; j < 3; j++) {
         cout << " " << board[i][j] << " ";
         if(j < 2) cout << "|";
      }
      cout << "\n";
      if(i < 2) cout << "-----------\n";
   }
}
bool MovesLeft() {
   for(int i=0;i<3;i++)
      for(int j=0;j<3;j++)
         if(board[i][j] != 'X' && board[i][j] != 'O')
            return true;
   return false;
}
int evaluate(){
   for(int i=0;i<3;i++) {
      if(board[i][0]==board[i][1] && board[i][1]==board[i][2]) {
         if(board[i][0]=='O') return +10;
         if(board[i][0]=='X') return -10;
      }
      if(board[0][i]==board[1][i] && board[1][i]==board[2][i]) {
         if(board[0][i]=='O') return +10;
         if(board[0][i]=='X') return -10;
      }
   }
   if(board[0][0]==board[1][1] && board[1][1]==board[2][2]) {
      if(board[0][0]=='O') return +10;
      if(board[0][0]=='X') return -10;
   }
   if(board[0][2]==board[1][1] && board[1][1]==board[2][0]) {
      if(board[0][2]=='O') return +10;
      if(board[0][2]=='X') return -10;
   }
   return 0;
}
int minimax(bool Ismax){
   int score = evaluate();
```

```
    if(score==10 || score == -10){
        return score;
    }
    if(!MovesLeft()){
        return 0;
    }
    if(Ismax){
        int best = -500;
                for(int i=0;i<3;i++) {
            for(int j=0;j<3;j++) {
                if(board[i][j] != 'X' && board[i][j] != 'O') {
                    char temp = board[i][j];
                    board[i][j] = 'O';
                    best = max(best, minimax(false));
                    board[i][j] = temp;
                }
            }
        }
        return best;
    }
        else {
        int best = 500;
        for(int i=0;i<3;i++) {
            for(int j=0;j<3;j++) {

                if(board[i][j] != 'X' && board[i][j] != 'O') {

                    char temp = board[i][j];
                    board[i][j] = 'X';

                    best = min(best, minimax(true));

                    board[i][j] = temp;
                }
            }
        }
        return best;
    }
}
int findBestMove() {
    int bestVal = -1000;
    int bestMove = -1;
    for(int i=0;i<3;i++) {
        for(int j=0;j<3;j++) {
            if(board[i][j] != 'X' && board[i][j] != 'O') {
                char temp = board[i][j];
                board[i][j] = 'O';
                int moveVal = minimax(false);
                board[i][j] = temp;
```

```
            if(moveVal > bestVal) {
                bestMove = temp - '0';
                bestVal = moveVal;
            }
        }
    }
  }
  return bestMove;
}
int main(){
  cout << "===== Tic Tac Toe Using MiniMax ====\n";
  while(true) {
    displayBoard();
    int choice;
    cout << "\nEnter your move (1-9): ";
    cin >> choice;
    int row = (choice-1)/3;
    int col = (choice-1)%3;
    if(board[row][col]=='X' || board[row][col]=='O') {
      cout << "Invalid move\n";
      continue;
    }
    board[row][col]='X';
    if(evaluate()==-10) {
    displayBoard();
    cout << "\nYou Win\n";
    break;
    }
    if(!MovesLeft()) {
      displayBoard();
      cout << "\nDraw\n";
      break;
    }
    int aiMove = findBestMove();
    int r = (aiMove-1)/3;
    int c = (aiMove-1)%3;
    board[r][c]='O';
    cout << "\nAI chose position " << aiMove << "\n";
    if(evaluate()==10) {
      displayBoard();
      cout << "\nAI Wins\n";
      break;
    }
    if(!MovesLeft()) {
      displayBoard();
      cout << "\nDraw\n";
      break;
    }
  }
```

```
    return 0;
}
```

```
Enter your move (1-9): 1

AI chose position 5

 X | 2 | 3
-----------
 4 | O | 6
-----------
 7 | 8 | 9

Enter your move (1-9): 9

AI chose position 2

 X | O | 3
-----------
 4 | O | 6
-----------
 7 | 8 | X

Enter your move (1-9): 8

AI chose position 7

 X | O | 3
-----------
 4 | O | 6
-----------
 O | X | X

Enter your move (1-9): 3

AI chose position 6

 X | O | X
-----------
 4 | O | O
-----------
 O | X | X

Enter your move (1-9): 4

 X | O | X
-----------
 X | O | O
-----------
 O | X | X

Draw
```